

FATIMA ABU BAKAR

Software Engineering Graduate • Front-End & AI-Integrated Web Development

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PROFESSIONAL SUMMARY

Final-year Software Engineering student at Lahore Garrison University with hands-on experience in front-end development and AI-powered web applications. Built AccessAble, an AI-driven web accessibility tool, and SoleCraft, a production-grade offline-first e-commerce storefront demonstrating strong command of modern web architecture. Fast learner with proven ability to deliver under pressure, actively seeking a Junior Developer or Software Engineer role (remote or Lahore-based).

EDUCATION

Bachelor of Software Engineering | Lahore Garrison University — DHA Phase 6, Lahore

Oct 2022 – Jun 2026

Expected Graduation: November 2026

- Completed 8 semesters of core CS and software engineering coursework
- Relevant coursework: Data Structures, OOP, Database Systems, Web Engineering, Software Design, AI

EXPERIENCE

Summer Internship | ISPR (Inter-Services Public Relations)

Jul 2025 – Aug 2025 (6 weeks)

- Collaborated with engineering teams on requirement gathering and system analysis
- Gained professional experience working within a large-scale government technology environment

PROJECTS

SoleCraft — Offline-First E-Commerce Storefront

Current

Next.js 15, React 19, TypeScript, Tailwind CSS — Personal Project

- Built a production-grade e-commerce storefront with offline-first architecture using Service Workers and IndexedDB
- Implemented a Web Worker-driven shopping cart with a typed message protocol for off-main-thread price calculations
- Developed a custom Service Worker handling offline request queuing, cache management, and auto-sync on connectivity restoration
- Created an interactive 3D product viewer using React Three Fiber for immersive product visualization
- Maintained strict TypeScript across the entire codebase, including Web Workers and service worker logic

AccessAble — Final Year Project

HTML, CSS, JavaScript, Python, AI/ML APIs

- Built an AI-powered web accessibility tool that helps users with disabilities navigate websites more easily
- Integrated AI features to auto-generate alt text, improve contrast, and adjust layouts for accessibility compliance
- Designed end-to-end: requirements, UML diagrams, system architecture, and front-end implementation
- Directly addresses WCAG 2.1 standards — a real-world, deployable solution

StudyPuff — Study Tracker Web App

HTML, CSS, JavaScript

- Developed a front-end productivity app for students to track study sessions and set goals
- Implemented local storage to persist user data across sessions without a backend

Interactive Mini-Games Suite

HTML, CSS, JavaScript — DOM Manipulation

- Built Rock Paper Scissors and Tic Tac Toe games with fully interactive UI and win/draw detection logic
- Demonstrates solid understanding of JavaScript events, state management, and DOM manipulation

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL, HTML5, CSS3

Front-End: Next.js, React (incl. React Three Fiber), Tailwind CSS, responsive design, DOM manipulation

Architecture: Service Workers, IndexedDB, Web Workers, offline-first design, REST APIs

Tools & IDEs: VS Code, Git & GitHub, Figma (basic), Draw.io, Lucidchart

Concepts: OOP, UML diagrams, SRS documentation, WCAG accessibility standards

AI/ML: API integration, accessibility automation (AccessAble FYP)

Languages Spoken: English (fluent — written & spoken), Urdu (native)

ACHIEVEMENTS & CERTIFICATIONS

- Completed ISPR internship and received official certificate of completion (2025)
- Delivered AccessAble FYP — AI-powered accessibility tool addressing real-world disability challenges
- Built SoleCraft, a self-directed, production-grade offline-first e-commerce application
- Consistently managed group projects and presentations throughout degree program